

KARTHIK S

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Education

B.Tech in Artificial Intelligence & Data Science

Sri Krishna College of Technology
2022 – 2026 | Coimbatore, Tamil Nadu
CGPA: 7.3 / 10

Technical Skills

Languages

Python, SQL, Java (Basic), HTML, CSS, JavaScript

Core

DBMS, OOP, CN, OS

AI / ML Frameworks

Scikit-learn, PyTorch, XGBoost, Random Forest, Linear Regression, HuggingFace Transformers, Transformer Architecture

Generative AI & LLMs

LangChain, LlamaIndex, RAG (Retrieval-Augmented Generation), LLM APIs, ChromaDB (Vector Databases), MLFlow

Data Science

Feature Engineering, Hyperparameter Tuning, Model Evaluation.

Web & API Dev

FastAPI, Flask, React.js, REST APIs, Streamlit, Modular Architecture, OpenAI

DevOps & Tools

PostgreSQL, Docker (Basic), Kubernetes, Azure, Git, Linux

RELEVANT COURSEWORK

Data Structures & Algorithms

Problem-solving with Python.

Machine Learning & Data Science

Supervised/unsupervised learning, model evaluation, and deployment pipelines

Deep Learning Fundamentals

Neural networks, CNNs, Transformer architecture, and optimization techniques

Database Management Systems

SQL, relational modeling, and query optimization

Certificates

Business Fundamentals for Entrepreneurs

issued by NPTEL

IoT Edge Computing and IoT Analytics

issued by Infosys Springboard

Professional Experience

Gramosoft Private Limited

Mar 2026 – Jun 2026

AI/ML Intern

Chennai, India

- Designed and implemented retrieval optimization for the company's internal document Q&A system using LangChain and ChromaDB. Tested multiple chunking strategies and embedding models to improve search relevance and retrieval accuracy.
- Optimized text preprocessing pipelines for 1,000+ internal documents (PDFs, Word files), handling extraction, cleaning, and structured chunking. Delivered processed data ahead of schedule to support team's integration timeline.
- Built an internal Streamlit dashboard for testing and validating retrieval results, enabling the team to quickly iterate on prompt engineering and configuration changes without writing additional code.
- Actively participated in company-led training sessions on LLM evaluation techniques, RAG architecture, and production deployment best practices. Completed internal assessments on prompt engineering and retrieval evaluation metrics.

Projects

AI Interview Simulation System

2026 – Present

- Architecting an end-to-end AI-powered interview simulation platform using LangChain and LlamaIndex to deliver dynamic, context-aware interview questions tailored to a candidate's resume and target role.
- Implementing a RAG pipeline backed by ChromaDB vector database to retrieve domain-specific knowledge and generate accurate, role-relevant follow-up questions in real time.
- Integrating HuggingFace Transformer models for candidate response evaluation, scoring communication quality, technical depth, and keyword alignment.
- Building a FastAPI backend with a Streamlit-based interactive UI, containerized using Docker for consistent cross-environment deployment.
- Tracking model experiments and evaluation metrics using MLflow to ensure reproducibility and iterative performance improvement.

Insurance Claims Anomaly Detection

Aug 2025 – Sep 2025

- Trained an XGBoost classification model to detect fraudulent insurance claims, achieving 94% accuracy on held-out test data.
- Applied advanced feature engineering and class-imbalance handling to improve minority-class detection and reduce false negatives.
- Evaluated model robustness using Precision, Recall, F1-Score, and ROC-AUC; generated performance reports for stakeholder communication.
- Reduced simulated manual claim review workload by ~30% through automated anomaly flagging in the claims pipeline.

Supermarket E-Commerce Web Application

Jul 2024 – Aug 2024

- Developed a responsive grocery e-commerce app using React.js with reusable component architecture, enabling seamless product browsing and cart management.
- Connected the application with RESTful APIs and database services to perform dynamic CRUD operations for products, users, and orders.
- Integrated MySQL database for managing product details, user information, and order data, ensuring efficient data storage and retrieval.